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WeBWorK assignment number WW_01 is due : 07/14/2008 at 11:00pm CDT.

The
(* replace with url for the course home page *)
for the course contains the syllabus, grading policy and other information.

This file is /conf/snippets/setHeader.pg you can use it as a model for creating files which introduce each problem set.

The primary purpose of WeBWorK is to let you know that you are getting the correct answer or to alert you if you are making some kind of mistake. Usually you can attempt a problem as many times as you want before the due date. However, if you are having trouble figuring out your error, you should consult the book, or ask a fellow student, one of the TA's or your professor for help. Don't spend a lot of time guessing – it's not very efficient or effective.

Give 4 or 5 significant digits for (floating point) numerical answers. For most problems when entering numerical answers, you can if you wish enter elementary expressions such as $2 \wedge 3$ instead of 8, $sin(3*pi/2)$instead of -1, $e \wedge (ln(2))$ instead of 2, $(2+tan(3))*(4-sin(5))\wedge 6-7/8$ instead of 27620.3413, etc. Here's the **list of the functions** which WeBWorK understands.

You can use the Feedback button on each problem page to send e-mail to the professors.

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1. **(1 pt) ttulib/M2360/chap1/leon_1_1/n01.pg**
(*S. 1-1*)
Use back substitution to solve the following systems. Enter your answers as points. (E.g., if the solution is $x_1 = 2, x_2 = -1$ you would enter

(2,-1)

Do not forget the parentheses and commas.)

(a) $\begin{array}{rcrcr} 2x_1 & - & x_2 & = & -7 \\ & & 3x_2 & = & 9 \end{array}$

$(x_1, x_2) =$ ______

(b) $\begin{array}{rcrcrcr} x_1 & + & 2x_2 & - & x_3 & = & -8 \\ & & 3x_2 & + & 3x_3 & = & -15 \\ & & & & x_3 & = & 0 \end{array}$

$(x_1, x_2, x_3) =$ ______

(c) $\begin{array}{rcrcrcrcr} 4x_1 & - & 2x_2 & + & x_3 & & & = & 0.5 \\ & & x_2 & & & - & 3x_4 & = & 4 \\ & & & & -2x_3 & + & 3x_4 & = & -8 \\ & & & & & & x_4 & = & -1 \end{array}$

$(x_1, x_2, x_3, x_4) =$ ______

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2. **(1 pt) ttulib/M2360/chap1/leon_1_1/n02.pg**
(*S. 1-1*)
Write the coefficient and augmented matrices for each of the following systems:

(a) $\begin{array}{rcrcr} 2x_1 & - & x_2 & = & -7 \\ & & 3x_2 & = & 9 \end{array}$

Coefficient matrix = $\begin{bmatrix} \_\_ & \_\_ \\ \_\_ & \_\_ \end{bmatrix}$

Augmented matrix = $\left[\begin{array}{cc|c} \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \end{array}\right]$

(b) $\begin{array}{rcrcrcr} x_1 & + & 2x_2 & - & x_3 & = & -8 \\ & & 3x_2 & + & 3x_3 & = & -15 \\ & & & & x_3 & = & 0 \end{array}$

Coefficient matrix = $\begin{bmatrix} \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \end{bmatrix}$

Augmented matrix = $\left[\begin{array}{ccc|c} \_\_ & \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ & \_\_ \end{array}\right]$

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3. **(1 pt) ttulib/M2360/chap1/leon_1_1/n03.pg**
(*S. 1-1*)
Solve each of the following systems:

(a) $\begin{array}{rcrcr} 2x_1 & - & x_2 & = & -7 \\ 4x_1 & + & 3x_2 & = & 1 \end{array}$

$(x_1, x_2) =$ ______

(b) $\begin{array}{rcrcrcr} x_1 & + & 3x_2 & - & x_3 & = & 9 \\ 4x_1 & - & x_2 & + & 2x_3 & = & -3 \\ 3x_1 & & & + & x_3 & = & 0 \end{array}$

$(x_1, x_2, x_3) =$ ______

(c) $\begin{array}{rcrcrcrcr} x_1 & - & x_2 & & & + & x_4 & = & 4.3 \\ 4x_1 & & & + & 2x_3 & & & = & 4.8 \\ 3x_1 & & & + & x_3 & + & x_4 & = & 4.6 \\ & & x_2 & + & 2x_3 & + & x_4 & = & -1.1 \end{array}$

$(x_1, x_2, x_3, x_4) =$ ______

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4. **(1 pt) ttulib/M2360/chap1/leon_1_2/n01.pg**
(*S. 1-2*)
Indicate which of the following are in row echelon form and which are in reduced row echelon form:

(a) $\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

Row echelon form?:

- yes
- no

Reduced row echelon form?:

- yes
- no

(b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Row echelon form?:

- yes
- no

Reduced row echelon form?:

- yes
- no

(c) $\begin{bmatrix} 1 & 3 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

Row echelon form?:

- yes
- no

Reduced row echelon form?:

- yes
- no

(d) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$

Row echelon form?:

- yes
- no

Reduced row echelon form?:

- yes
- no

(e) $\begin{bmatrix} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -6 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$

Row echelon form?:

- yes
- no

Reduced row echelon form?:

- yes
- no

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5. **(1 pt) ttulib/M2360/chap1/leon_1_2/n02.pg**

In each of the following, the augmented matrix is in row echelon form. Indicate whether the corresponding linear system is consistent. If the system has a unique solution, find it.

(a) $\left[\begin{array}{cc|c} 1 & 3 & 1 \\ 0 & 1 & 2 \end{array}\right]$

consistent?

- yes
- no

solution (if none, enter “NONE”): $(x_1, x_2) =$ ____________

(b) $\left[\begin{array}{cc|c} 1 & -3 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array}\right]$

consistent?

- yes
- no

solution (if none, enter “NONE”): $(x_1, x_2) =$ ____________

(c) $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right]$

consistent?

- yes
- no

solution (if none, enter “NONE”): $(x_1, x_2, x_3) =$ ____________

(d) $\left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{array}\right]$

consistent?

- yes
- no

solution (if none, enter “NONE”): $(x_1, x_2, x_3) =$ ____________

6. **(1 pt) ttulib/M2360/chap1/leon_1_2/n03.pg**
(*S. 1-2*)
In each of the following, the augmented matrix is in reduced row echelon form. Find the solution set to the corresponding linear system.
(Example: Suppose a system has as its solution set the set of all 5-tuples $(1-\alpha-\beta, \alpha, \beta, 2, -1)$, where α, β range over the real numbers. You would enter this solution as `(1 - a - b, a, b, 2, -1)`, using as your variables any of the letters `a, b, c, ... ,t .`
If the system has no solutions, enter `NONE`.)

(a) $\left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right]$
Solution: $(x_1, x_2) =$ ______

(b) $\left[\begin{array}{ccc|c} 1 & -3 & 0 & 2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right]$
Solution: $(x_1, x_2, x_3) =$ ______

(c) $\left[\begin{array}{cccc|c} 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$
Solution: $(x_1, x_2, x_3, x_4) =$ ______

(d) $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$
Solution: $(x_1, x_2, x_3) =$ ______

7. **(1 pt) ttulib/M2360/chap1/leon_1_2/n06.pg**
(*S. 1-2, cf. no. 5*)
Attempt to solve the following linear systems by using Gaussian elimination to find a similar system whose coefficient matrix is in row echelon form.

If the system has a unique solution, give it (in point form.)

If the system is inconsistent, write `NONE` for the solutions.

If the system has is consistent and there are free variables, transform it to reduced row echelon form and find all solutions. (As before, write your general solutions in terms of the variables `a, b, c, ...`. For example, `(1,2-a+b,b,a)`.)

(a) $\begin{array}{rcrcr} 2x_1 & - & x_2 & = & -7 \\ x_1 & + & 3x_2 & = & 7 \end{array}$
$(x_1, x_2) =$ ______

(b) $\begin{array}{rcrcrcr} x_1 & + & 3x_2 & + & 2x_3 & = & 1 \\ 2x_1 & + & x_2 & - & x_3 & = & 2 \end{array}$
$(x_1, x_2, x_3) =$ ______

(c) $\begin{array}{rcrcrcr} x_1 & + & 2x_2 & + & 3x_3 & = & 1 \\ 2x_1 & + & x_2 & & & = & 2 \\ 4x_1 & + & 5x_2 & + & 6x_3 & = & 1 \end{array}$
$(x_1, x_2, x_3) =$ ______

(d) $\begin{array}{rcrcrcrcr} 2x_1 & + & x_2 & + & 4x_3 & + & 3x_4 & = & 15 \\ x_1 & & & + & x_3 & + & x_4 & = & 5 \\ & & x_2 & + & x_3 & + & 2x_4 & = & 6 \\ 2x_1 & - & 3x_2 & + & x_3 & + & x_4 & = & 4 \end{array}$
$(x_1, x_2, x_3, x_4) =$ ______

(e) $\begin{array}{rcrcrcrcr} 2x_1 & + & x_2 & + & 4x_3 & + & 3x_4 & = & 15 \\ x_1 & & & + & x_3 & + & x_4 & = & 5 \\ & & x_2 & + & x_3 & + & 2x_4 & = & 6 \\ 3x_1 & + & 3x_2 & + & 8x_3 & + & 7x_4 & = & 31 \end{array}$
$(x_1, x_2, x_3, x_4) =$ ______

(f) $\begin{array}{rcrcrcrcr} 2x_1 & + & x_2 & - & x_3 & + & 3x_4 & = & 3 \\ x_1 & + & 2x_2 & & & + & x_4 & = & 2 \end{array}$
$(x_1, x_2, x_3, x_4) =$ ______

8. **(1 pt) ttulib/M2360/chap1/leon_1_2/ur_la_1_6.pg**

For each system, determine whether it has a unique solution (in this case, find the solution), infinitely many solutions, or no solutions.

1.
$$\begin{cases} -7x+8y=-75 \\ 4x+5y=-5 \end{cases}$$

- A. Infinitely many solutions
- B. Unique solution: $x = 5,\ y = -5$
- C. Unique solution: $x = -5,\ y = 5$
- D. Unique solution: $x = 0,\ y = 0$
- E. No solutions
- F. None of the above

2.
$$\begin{cases} 4x+8y=0 \\ -3x-7y=0 \end{cases}$$

- A. No solutions
- B. Unique solution: $x = 0,\ y = 0$
- C. Unique solution: $x = 12,\ y = -10$
- D. Unique solution: $x = 7,\ y = 4$
- E. Infinitely many solutions
- F. None of the above

3.
$$\begin{cases} -2x+3y=12 \\ -6x+9y=36 \end{cases}$$

- A. No solutions
- B. Unique solution: $x = 12,\ y = 36$
- C. Unique solution: $x = 0,\ y = 0$
- D. Unique solution: $x = -6,\ y = 0$

- E. Infinitely many solutions
- F. None of the above

4.

$$\begin{cases} 4x+2y=-16 \\ 12x+6y=-47 \end{cases}$$

- A. Unique solution: $x=-47,\ y=-16$
- B. Unique solution: $x=0,\ y=0$
- C. No solutions
- D. Unique solution: $x=-16,\ y=-47$
- E. Infinitely many solutions
- F. None of the above

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9. **(1 pt) ttulib/M2360/chap1/leon_1_2/ur_la_1_22.pg**

The reduced row-echelon forms of the augmented matrices of four systems are given below. How many solutions does each system have?

1. $\left[\begin{array}{cc|c} 1 & 0 & 10 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{array}\right]$

- A. Infinitely many solutions
- B. Unique solution
- C. No solutions
- D. None of the above

2. $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -7 \end{array}\right]$

- A. Infinitely many solutions
- B. No solutions
- C. Unique solution
- D. None of the above

3. $\left[\begin{array}{ccc|c} 1 & 0 & 11 & 0 \\ 0 & 1 & 16 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$

- A. No solutions
- B. Unique solution
- C. Infinitely many solutions
- D. None of the above

4. $\left[\begin{array}{ccc|c} 1 & 0 & -8 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$

- A. Unique solution
- B. No solutions
- C. Infinitely many solutions
- D. None of the above

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10. **(1 pt) ttulib/M2360/chap1/leon_1_3/n01.pg**

(*S. 1-3, cf. no. 1)*)

Let $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 2 \\ -2 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 & 2 \\ -3 & 1 & 0 \\ 2 & -4 & 1 \end{bmatrix}$.

Compute (remember how we enter matrices!):

(a) $2A =$

(b) $A-3B =$

(c) $A^T-(3B)^T =$

(d) $AB =$

(e) $BA =$

(f) $A^TB =$

(g) $(AB)^T =$

(g) $(B^TA^T) =$

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11. **(1 pt) ttulib/M2360/chap1/leon_1_3/n02.pg**

(*S. 1-3, cf. no. 2)*)

For each of the following pairs of matrices, determine whether it is possible to multiply the first times the second.

If it is possible, write the product.

Write `undefined` if it is not possible. (Make sure you get the spelling correct!)

(a) $\begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 3 & 0 \\ -1 & 1 \end{bmatrix} =$

(b) $\begin{bmatrix} 0 & 1 \\ 3 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 0 \end{bmatrix} =$

(c) $\begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} -2 & 5 \end{bmatrix} =$

(d) $\begin{bmatrix} -2 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} =$

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12. **(1 pt) ttulib/M2360/chap1/leon_1_3/ur_Ch1_3_4.pg**

$5x+4y-3z=9$
$-5x-8y+8z=7$
$-9x-2y-6z=6$

Write the above system of equations in matrix form:

$$\begin{bmatrix} \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \end{bmatrix} * \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \_\_ \\ \_\_ \\ \_\_ \end{bmatrix}$$

13. **(1 pt) ttulib/M2360/chap1/leon_1_3/ur_la_3_1.pg**

If $A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 1 \\ -2 & -3 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & 0 & 2 \\ 1 & 0 & 1 \\ 0 & 3 & 0 \end{bmatrix}$

Then $4A+B = \begin{bmatrix} \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \end{bmatrix}$

and $A^T = \begin{bmatrix} \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \end{bmatrix}$

14. **(1 pt) ttulib/M2360/chap1/leon_1_3/ur_la_3_2.pg**

If $A = \begin{bmatrix} 1 & 4 & 2 \\ 0 & 4 & 3 \\ -3 & 2 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & 0 & -3 \\ 1 & -3 & 4 \\ -2 & 4 & 0 \end{bmatrix}$

Then $AB = \begin{bmatrix} \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \end{bmatrix}$

and $BA = \begin{bmatrix} \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \end{bmatrix}$

15. **(1 pt) ttulib/M2360/chap1/leon_1_3/ur_la_3_4.pg**

Compute the following product.

$$\begin{bmatrix} -4 & -1 \\ -4 & -2 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -3 & -3 \end{bmatrix} = \begin{bmatrix} \_\_\_ & \_\_\_ \\ \_\_\_ & \_\_\_ \\ \_\_\_ & \_\_\_ \end{bmatrix}$$

16. **(1 pt) ttulib/M2360/chap1/leon_1_3/ur_la_3_6.pg**

If A and B are 3×2 matrices, and C is a 7×3 matrix, which of the following are defined?

- A. $A+B$
- B. $B^T C^T$
- C. CA
- D. $C+B$
- E. AB
- F. A^T

17. **(1 pt) ttulib/M2360/chap1/leon_1_3/ur_la_3_13.pg**

If A, B, and C are 2×2, 2×4, and 4×7 matrices respectively, determine which of the following products are defined. For those defined, enter the size of the resulting matrix (e.g. "3 x 4", with spaces between numbers and "x"). For those undefined, enter "undefined".

CB: _______
BC: _______
AB: _______
AC: _______

18. **(1 pt) ttulib/M2360/chap1/leon_1_3/ur_la_3_18.pg**

Compute the following product.

$$\begin{bmatrix} 1 & 0 & 5 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 5 \\ -4 \end{bmatrix} = \begin{bmatrix} \_\_\_\_\_ \end{bmatrix}$$

19. **(1 pt) ttulib/M2360/chap1/leon_1_3/ur_la_3_19.pg**

Compute the following product.

$$\begin{bmatrix} 5 & 4 & 5 \\ -3 & 5 & -1 \\ 5 & 4 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} \_\_\_ \\ \_\_\_ \\ \_\_\_ \end{bmatrix}$$

20. **(1 pt) ttulib/M2360/chap1/leon_1_3/ur_la_3_20.pg**

Find a 3×3 matrix A such that

$A \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ -4 \\ -3 \end{bmatrix}$,

$A \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -5 \end{bmatrix}$, and

$A \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \\ 4 \end{bmatrix}$.

$A = \begin{bmatrix} \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \end{bmatrix}$

21. **(1 pt) ttulib/M2360/chap1/leon_1_3/ur_la_3_30.pg**

Reduce the matrix $\begin{bmatrix} 2 & 3 & -13 \\ 1 & -1 & 1 \end{bmatrix}$ to reduced row-echelon form.

$$\begin{bmatrix} \_\_\_ & \_\_\_ & \_\_\_ \\ \_\_\_ & \_\_\_ & \_\_\_ \end{bmatrix}$$

22. **(1 pt) ttulib/M2360/chap1/leon_1_3/ur_la_3_34.pg**

Find a non-zero, two-by-two matrix such that:

$A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ $A = \begin{bmatrix} \_\_ & \_\_ \\ \_\_ & \_\_ \end{bmatrix}$

23. **(1 pt) ttulib/M2360/chap1/leon_1_4/n03.pg**

(*S. 1-4, cf. no. 3)*)

For each of the following pairs of matrices, find an elementary matrix E such that $EA = B$. (Notice that B can be obtained from A by a single row operation.)

(a) $A = \begin{bmatrix} 1 & 3 \\ -2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -2 & 1 \\ 1 & 3 \end{bmatrix}$

$E =$____________________

(b) $A = \begin{bmatrix} 1 & 3 & 0 \\ 1 & 1 & 1 \\ 5 & 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 & 0 \\ -4 & -2 & -1 \\ 5 & 3 & 2 \end{bmatrix}$

$E =$____________________

(c) $A = \begin{bmatrix} 8 & 1 & 0 \\ 3 & 1 & 5 \\ 2 & 4 & 8 \end{bmatrix}, B = \begin{bmatrix} 8 & 1 & 0 \\ 3 & 1 & 5 \\ 1 & 2 & 4 \end{bmatrix}$

$E =$ ____________________

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24. (1 pt) ttulib/M2360/chap1/leon_1_4/n08.pg
(*S. 1-4, cf. no. 8)*)

Let

$$A = \begin{bmatrix} 3 & 1 & 2 \\ 2 & 1 & 1 \\ -2 & -1 & 0 \end{bmatrix}.$$

Verify that

$$A^{-1} = \begin{bmatrix} 1 & -2 & -1 \\ -2 & 4 & 1 \\ 0 & 1 & 1 \end{bmatrix},$$

and use it to solve the system $A\mathbf{x} = \mathbf{b}$ for the following choices of $\mathbf{b}$:

(a) $\mathbf{b} = \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}^T$

$(x_1, x_2, x_3) =$ ____________________

(Notice that you should enter a point for your answer.)

(b) $\mathbf{b} = \begin{bmatrix} 1 & -2 & 4 \end{bmatrix}^T$

$(x_1, x_2, x_3) =$ ____________________

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25. (1 pt) ttulib/M2360/chap1/leon_1_4/n09.pg
(*S. 1-4, cf. no. 9)*)

Find the inverse of each of the following matrices:

(a) $A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}, \quad A^{-1} =$ ____________________

(b) $A = \begin{bmatrix} 1 & 2 & 1 \\ -2 & 1 & 1 \\ 3 & 5 & 2 \end{bmatrix}, \quad A^{-1} =$ ____________________

(c) $A = \begin{bmatrix} 1 & -1 & -1 \\ 2 & 1 & 2 \\ 0 & 1 & 0 \end{bmatrix}, \quad A^{-1} =$ ____________________

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26. (1 pt) ttulib/M2360/chap1/leon_1_4/ur_Ch2_1_4.pg

Are the following matrices invertible? Enter "Y" or "N".
You must get all of the answers correct to receive credit.

___1. $\begin{bmatrix} -6 & -6 \\ 1 & 8 \end{bmatrix}$

___2. $\begin{bmatrix} 4 & 7 \\ 0 & 0 \end{bmatrix}$

___3. $\begin{bmatrix} -2 & 7 \\ 4 & -14 \end{bmatrix}$

___4. $\begin{bmatrix} 6 & -2 \\ 1 & 9 \end{bmatrix}$

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27. (1 pt) ttulib/M2360/chap1/leon_1_4/ur_la_4_1.pg

If $A = \begin{bmatrix} -1 & 2 \\ 3 & 9 \end{bmatrix}$

Then $A^{-1} = \begin{bmatrix} \_\_ & \_\_ \\ \_\_ & \_\_ \end{bmatrix}$

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28. (1 pt) ttulib/M2360/chap1/leon_1_4/ur_la_4_2.pg

The matrix $\begin{bmatrix} 2 & 2 \\ -7 & k \end{bmatrix}$ is invertible if and only if $k \neq$ ___.

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29. (1 pt) ttulib/M2360/chap1/leon_1_4/ur_la_4_3.pg

If $A = \begin{bmatrix} 1 & 3 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}$

Then $A^{-1} = \begin{bmatrix} \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \end{bmatrix}$

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30. (1 pt) ttulib/M2360/chap1/leon_1_4/ur_la_4_5.pg

A square matrix is called a permutation matrix if it contains the entry 1 exactly once in each row and in each column, with all other entries being 0. All permutation matrices are invertible. Find the inverse of the following permutation matrix

$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

$A^{-1} = \begin{bmatrix} \_\_ & \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ & \_\_ \end{bmatrix}$

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31. (1 pt) ttulib/M2360/chap1/leon_1_2/n04.pg
(*S. 1-2*)

WebWork uses a notation for entering matrices that is similar to that used by a number of other computer programs and calculators. (It is very similar to a method for entering matrices in your TI-86 calculators for example.)

For example, to enter the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, you would just enter

```
[ [1, 2], [3, 4] ]
```

Go ahead, try it: __________

For convenience, you will sometimes be supplied with a multi-line answer box for your answer. WebWork does not care where you insert line breaks (as long as it is not in the middle of a number or variable name!), so you can enter $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ as

```
[ [1, 2],
  [3, 4] ]
```

putting different rows on different lines. (This helps in proofreading your answers.) **Remember to put commas between the rows,** or WebWork will think you are trying to multiply the rows.

Now for some serious questions. Enter the *reduced* row echelon form of the following matrices. Approximate answers should be entered to at least 4 decimal places. (You can also enter fractions such as 5/4.):

(a) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 9 \end{bmatrix}$

(c) $\begin{bmatrix} 3 & 1 & 0 & 4 \\ 1 & 5 & 2 & 1 \\ 2 & 2 & 4 & 2 \\ 2 & -4 & -2 & 3 \\ 3 & 7 & 6 & 3 \end{bmatrix}$

---

32. (1 pt) ttulib/M2360/chap1/leon_1_2/n05.pg
(*S. 1-2*)

Enter the *reduced* row echelon form of the following matrix, which contains a variable x. (Your answer should contain the variable x as well.

$$\begin{pmatrix} 2 & 1 & 3 \\ 1 & 2 & x \end{pmatrix}$$

---

33. (1 pt) ttulib/M2360/chap1/leon_1_2/ur_la_1_8.pg
Determine the value of h such that the matrix is the augmented matrix of a consistent linear system.

$$\left[\begin{array}{cc|c} 4 & -3 & h \\ -8 & 6 & 8 \end{array}\right]$$

$h =$ ____

---

34. (1 pt) ttulib/M2360/chap1/leon_1_2/ur_la_1_9.pg
Determine the value of h such that the matrix is the augmented matrix of a linear system with infinitely many solutions.

$$\left[\begin{array}{cc|c} 5 & -5 & 8 \\ 15 & h & 24 \end{array}\right]$$

$h =$ ____

---

35. (1 pt) ttulib/M2360/chap1/leon_1_2/ur_la_1_13.pg
Solve the system

$$\begin{cases} -4x_1 + x_2 = 5 \\ -12x_1 + 3x_2 = 15 \end{cases}$$

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \_\_ \\ \_\_ \end{bmatrix} + \begin{bmatrix} \_\_ \\ \_\_ \end{bmatrix} s.$

---

36. (1 pt) ttulib/M2360/chap1/leon_1_2/ur_la_1_23.pg
Write the system

$$\begin{cases} 11y - 6z = -7 \\ -5x - 2y = -4 \\ 9x + 3y + 10z = 4 \end{cases}$$

in matrix form.

$\begin{bmatrix} \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \_\_ \\ \_\_ \\ \_\_ \end{bmatrix}$

---

37. (1 pt) ttulib/M2360/chap1/leon_1_3/ur_la_3_1a.pg
If $A = \begin{bmatrix} -1 & -3 & 3 \\ -1 & 0 & 1 \\ 4 & 1 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 & 4 \\ -2 & -2 & 2 \\ 0 & -3 & 0 \end{bmatrix}$

Then $3A - 4B = \begin{bmatrix} \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \end{bmatrix}$

and $4A^T = \begin{bmatrix} \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \\ \_\_ & \_\_ & \_\_ \end{bmatrix}$

---

38. (1 pt) ttulib/M2360/chap1/leon_1_3/ur_la_3_12.pg
Compute the following product.

$\begin{bmatrix} 2 & -2 & -4 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ 3 & 2 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} \_\_\_\_ & \_\_\_\_ \end{bmatrix}$

---

39. (1 pt) ttulib/M2360/chap1/leon_1_3/ur_la_3_17.pg
Compute the following product.

$\begin{bmatrix} 5 & -4 & 1 \\ -3 & -2 & -3 \end{bmatrix} \begin{bmatrix} -4 \\ -3 \\ 1 \end{bmatrix} = \begin{bmatrix} \_\_\_\_ \\ \_\_\_\_ \end{bmatrix}$

---

40. (1 pt) ttulib/M2360/chap1/leon_1_3/ur_la_3_33.pg
Find a non-zero, two-by-two matrix such that:

$\begin{bmatrix} -8 & 4 \\ 40 & -20 \end{bmatrix} * \begin{bmatrix} \_\_ & \_\_ \\ \_\_ & \_\_ \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

---

41. (1 pt) ttulib/M2360/chap1/leon_1_3/ur_la_3_35.pg
Find a two-by-two matrix such that:

$\begin{bmatrix} -4 & 0 \\ -2 & 4 \end{bmatrix} * \begin{bmatrix} \_\_ & \_\_ \\ \_\_ & \_\_ \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$